CHAPTER 2

TECHNOLOGICAL NETWORKS

A discussion of engineered networks like the Internet and the power grid and how we determine their structure

IN THE next four chapters we define and describe some of the most commonly studied networks, dividing them into four general classes—technological networks, social networks, information networks, and biological networks. We will list the most important examples in each class and then describe the techniques used to measure their structure. (The classes are not rigorously defined and there is, as we will see, some overlap between them, with some networks belonging to more than one class. Nonetheless, the division into classes is a useful one, since networks in the same class are often treated using similar techniques or ideas.)

It is not our intention in this book to study any one network in great detail. Plenty of other books exist that do that. Nonetheless, network science is concerned with understanding and modeling the behavior of real-world networked systems and observational data are the starting point for essentially all the developments of the field, so the reader will find it useful to have a grasp of the types of data that are available, their strengths and limitations, and the means used to acquire them. In this chapter we look at technological networks, the physical infrastructure networks that have grown up over the last century or so and form the backbone of modern technological societies. Perhaps the most celebrated such network—and a relatively recent entry in the field—is the Internet, the global network of data connections, electrical, optical, and wireless, that links computers and other information systems together. Section 2.1 is devoted to a discussion of the Internet. A number of other important examples of technological networks, including power grids, transportation networks, delivery and distribution networks, and telephone networks, are discussed in subsequent sections.

2.1 THE INTERNET

The Internet should not be confused with the World Wide Web, a virtual network of web pages and hyperlinks, which we discuss in Section 4.1.

The Internet is the worldwide network of physical data connections between computers and related devices. The Internet is a *packet switched* data network, meaning that messages sent over it are broken up into *packets*, small chunks of data, that are sent separately over the network and reassembled into a complete message again at the other end. The format of the packets follows a standard known as the *Internet Protocol* (IP) and includes an *IP address* in each packet that specifies the packet's destination, so that it can be routed correctly across the network.

The telephone network is discussed in Section 2.2.

The alternative to a packet switched network is a *circuit switched* network, the classic example of which is the telephone system. In a circuit switched network, vertices request connections when needed, such as when a telephone call is placed, and the network allocates a separate circuit for each connection, reserved for the sole use of that connection until the connection is ended. This works well for voice traffic, which consists of discrete phone calls each with a definite beginning and end, but it would be a poor model for a data network, in which data transmission typically occurs in brief, intermittent bursts. Using a packet switched model for the Internet allows computers to transmit and receive data intermittently or at varying rates without tying up capacity on the network. By making packets reasonably small, we also allow for a certain amount of unreliability in the network. It is not uncommon for packets to disappear on the Internet and never reach their destination, sometimes because of hardware or software failure, but more often because packets are deliberately deleted to reduce congestion in the busiest parts of the network. If a message is divided into several packets before transmission and a few packets are lost, then only those that are lost need be resent to complete the message. A software protocol called *Transport Control Protocol* or TCP, which runs on top of IP, performs the necessary error checking and retransmission automatically, without the need for intervention from computer users or other software.[4]

The simplest network representation of the Internet (there are others, as we will shortly see) is one in which the vertices of the network represent computers and other devices, and the edges represent physical connections between them, such as optical fiber lines. In fact, ordinary computers mostly occupy only the vertices on the "outside" of the network, those that data flows to and from, but they do not act as intermediate points for the flow of data between others. (Indeed, most computers only have a single connection to the net, so it would not be possible for them to lie on the path between any others.) The "interior" nodes of the Internet are primarily *routers*, powerful special-purpose computers at the junctions between data lines that receive data packets and forward them in one direction or another towards their intended destination.

The general overall shape of the Internet is shown, in schematic form, in Fig. 2.1. The network is composed of three levels or circles of vertices. The innermost circle, the core of the network, is the *backbone* of the network, the trunk lines that provide long-distance high-bandwidth data transport across the globe, along with the high-performance routers and switching centers that link them together. These trunk lines are the highways of the Internet, built with the fastest fiber optic connections available (and improving all the time). The backbone is operated by *network backbone providers* (NBPs), who are primarily national governments and communications companies such

as AT&T, Global Crossing, British Telecom, and others.

The second circle of the Internet is composed of *Internet service providers* or ISPs—commercial companies, governments, universities, and others who contract with NBPs for connection to the backbone and then resell or otherwise provide that connection to end users, the ultimate consumers of Internet bandwidth, who form the third circle—businesses, government offices, academics, people in their homes, and so forth. In fact, as Fig. 2.1 shows, the ISPs are further subdivided into *regional ISPs* and *local* or *consumer ISPs*, the former being larger organizations whose primary customers are the local ISPs, who in turn sell network connections to the end users. This distinction is somewhat blurred however, because large consumer ISPs, such as America Online or British Telecom, often act as their own regional ISPs (and some may be backbone providers as well).

The network structure of the Internet is not dictated by any central authority. Protocols and guidelines are developed by an informal volunteer organization called the Internet Engineering Task Force, but one does not have to apply to any central Internet authority for permission to build a new spur on the Internet, or to take one out of service.

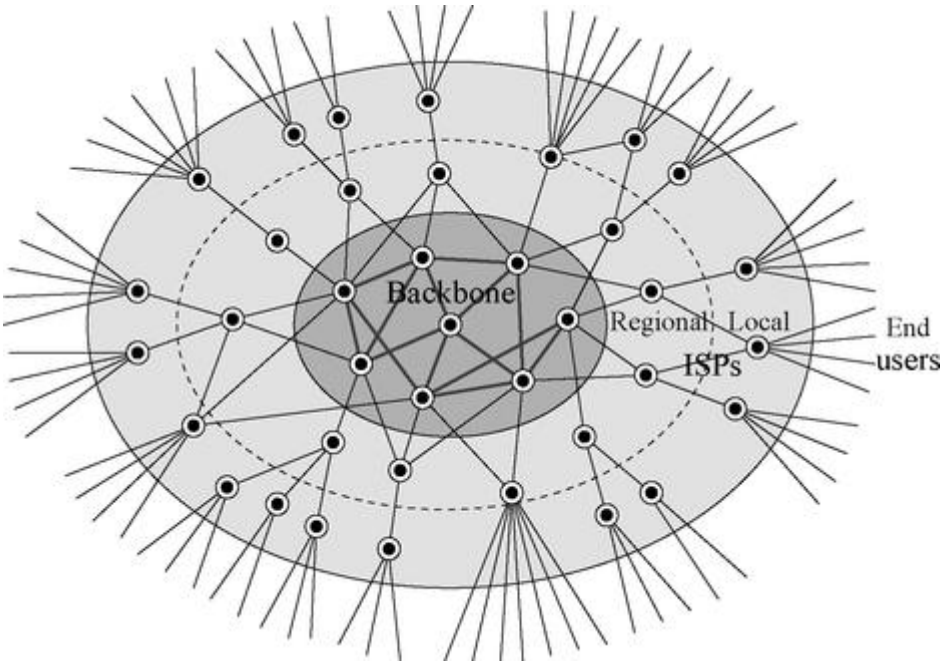


Figure 2.1: A schematic depiction of the structure of the Internet. The vertices and edges of the Internet fall into a number of different classes: the "backbone" of high-bandwidth long-distance connections; the ISPs, who connect to the backbone and who are divided roughly into regional (larger) and local (smaller) ISPs; and the end users—home users, companies, and so forth—who connect to the ISPs.

One of the remarkable features of the Internet is that the scheme used for the routing of packets from one destination to another is arrived at by automated negotiation among Internet routers using a system called the *Border Gateway Protocol* (BGP). BGP is designed in such a way that if new vertices or edges are added to the network, old ones disappear, or existing ones fail either permanently or temporarily, routers will take note and adjust their routing policy appropriately. Some human oversight is required to keep the system running smoothly, but no "Internet government" is needed to steer things from on high; the system organizes itself by the combined actions of many local and essentially autonomous computer systems.

While this is an excellent feature of the system from the point of view of robustness and flexibility, it is a problem for those who want to study the structure of the Internet, because there is

no central registry from which one can determine that structure. There is no one whose job it is to maintain an official map of the network. Instead the network's structure must be determined by experimental measurements. There are two primary methods for doing this. The first uses something called "traceroute"; the second uses BGP.

2.1.1 MEASURING INTERNET STRUCTURE USING TRACEROUTE

It is not, at least for most of us, possible to probe the network structure of the Internet directly. We can, however, quite easily discover the particular path taken by data packets traveling between our own computer (or any computer to which we have access) and most others on the Internet. The standard tool for doing this is called *traceroute*.

In addition to a destination address, which says where it is going, each Internet packet also contains a source address, which says where it started from, and a *time-to-live* (TTL). The TTL is a number that specifies the maximum number of "hops" that the packet can make to get to its destination, a hop being the traversal of one edge in the network. At every hop, the TTL is decreased by one, and if ever it reaches zero the packet is discarded, meaning it is deleted and not forwarded any further over the network. If we are using TCP, a message is also then sent back to the sender informing them that the packet was discarded and where it got to. (This is a part of TCP's mechanism for guaranteeing the reliable transmission of data—see above.) The TTL exists mainly as a safeguard to prevent packets from getting lost on the Internet and wandering around forever, but we can make use of it to track packet progress as well. The idea is as follows.

First, we send out a TCP packet with the destination address of the network vertex we are interested in and a TTL of 1. The packet makes a single hop to the first router along the way, its TTL is decreased to zero, the packet is discarded by the router and a message is returned to us telling us, among other things, the IP address of the router. We record this address and then repeat the process with a TTL of 2. This time the packet makes two hops before dying and the returned message tells us the IP address of the second router. The process is repeated with larger and larger TTL until the destination is reached, and the set of IP addresses received as a result specifies the entire route taken to get there.[5] There are standard software tools that will perform the entire procedure automatically and print out the list of IP addresses for us. On most computers the tool that does this is called "traceroute."

We can use traceroute (or a similar tool) to probe the network structure of the Internet. The idea is to assemble a large data set of traceroute paths between many different pairs of points on the Internet. With luck, most of the edges in the network (though usually not all of them) will appear at least once in this set, and the union of all of them should give a reasonably complete picture of the network. Early studies, for the sake of expediency, limited themselves to just a few source computers, but more recent ones, such as the DIMES Project,[6] make use of distributed collections of thousands of sources to develop a very complete picture of the network.

See Section 6.7 for a discussion of tree networks.

The paths from any single source to a set of destinations form a tree-like structure as shown schematically in Fig. 2.2a, b, and c.[7] The source computers should, ideally, be well distributed over the network. If they are close together, then there may be a substantial overlap between the traceroute paths to distant vertices, which means that they will duplicate needlessly each other's efforts, rather than returning independent measurements.

Once one has a suitable set of traceroute data, a simple union of all the paths appearing in the data set gives us our snapshot of the network structure—see Fig. 2.2d. That is, we go through each path and record a vertex for every IP address that appears in the path and an edge between every pair of addresses that appear in adjacent positions. As hinted above, it is unlikely that such a procedure will find all the edges in the network (see Fig. 2.2d again), and for studies based on small numbers of sources there can be quite severe biases in the sampling of edges [3,192]. However, better and better data sets are becoming available as time passes, and it is believed that we now have a reasonably complete picture of the shape of the Internet.

In fact, it is rarely, if ever, done to record *every* IP address on the Internet as a separate vertex. There are believed to be about 2 billion unique IP addresses in use on the Internet at any one time, with many of those corresponding to end-user computers that appear and disappear as the computers are turned on or off or connections to the Internet are made or broken. Most studies of the Internet ignore end-user computers and restrict themselves to just the routers, in effect concentrating on the inner zones in Fig. 2.1 and ignoring the outermost one. We will refer to such maps of the Internet as representations at the *router level*. The vertices in the network are routers, and the edges between them are network connections.

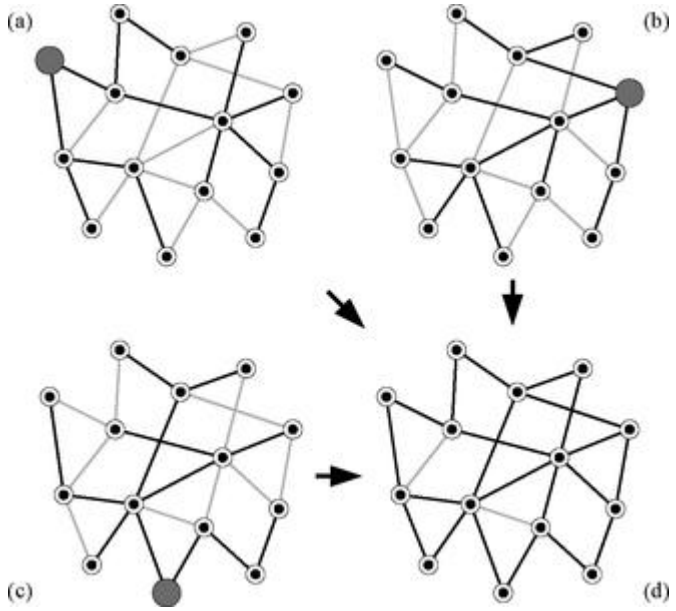


Figure 2.2: Reconstruction of the topology of the Internet from traceroute data. In panels (a), (b), and (c) we show in bold the edges in three sets of traceroute paths starting from each of the three highlighted source vertices. In panel (d) we form the union of these edges to make a picture of the overall network topology. Note that a few edges are missing from this picture (the remaining gray edges in panel (d)) because, by chance, they happen not to appear in any of the three individual traceroute data sets.

It may appear strange to ignore end-user computers, since the end users are, after all, the entire reason for the Internet's existence in the first place. However, it is the structure of the network at the router level that is responsible for most aspects of the performance, robustness, and efficiency of the network, that dictates the patterns of traffic flow on the network, and that forms the focus of most work on Internet structure and design. To the extent that these are the issues of scientific interest, therefore, it makes sense to concentrate our efforts on the router-level structure.

An example of a study of the topology of the Internet at the router level is that of Faloutsos *et al.* [111], who looked at the "degree distribution" of the network and discovered it to follow, approximately, a power law. We discuss degree distributions and power laws in networks in more detail in Section 8.4.

Even after removing all or most end-user computers from the network, the network structure at the router level may still be too detailed for our purposes. Often we would like a more coarse-grained representation of the network that gives us a broader overall picture of network structure. Such representations are created by grouping sets of IP addresses together into single vertices. Three different ways of grouping addresses are in common use giving rise to three different coarse-grained representations, at the level of subnets, domains, and autonomous systems.

A *subnet* is a group of IP addresses defined as follows. IP addresses consist of four numbers,

each one in the range from 0 to 255 (eight bits in binary) and typically written in a string separated by periods or dots. For example, the IP address of the main web server at the author's home institution, the University of Michigan, is 141.211.144.190. IP addresses are allocated to organizations in blocks. The University of Michigan, for instance, owns (among others) all the addresses of the form 141.211.144.xxx, where "xxx" can be any number between 0 and 255. Such a block, where the first three numbers in the address are fixed and the last can be anything, is called a *class C subnet*. There are also class B subnets, which have the form 141.211.xxx.yyy, and class A subnets, which have the form 141.xxx.yyy.zzz.

Since all the addresses in a class C subnet are usually allocated to the same organization, a reasonable way of coarse-graining Internet network data is to group vertices into class C subnets. In most cases this will group together vertices in the same organization, although larger organizations, like the University of Michigan, own more than one class C subnet, so there will still be more than one vertex in the coarse-grained network corresponding to such organizations. Given the topology of the network at the router level, the level of individual IP addresses, it is easy to lump together into a single vertex all addresses in each class C subnet and place an edge between any two subnets if any router in one has a network connection to any router in the other. Figure 1.1 on page 4 shows an example of the network structure of the Internet represented at the level of class C subnets.

The second common type of coarse-graining is coarse-graining at the domain level. A *domain* is a group of computers and routers under, usually, the control of a single organization and identified by a single *domain name*, normally the last two or three parts of a computer's address when the address is written in human-readable text form (as opposed to the raw IP addresses considered above). For example, "umich.edu" is the domain name for the University of Michigan and "oup . co . uk" is the domain name for Oxford University Press. The name of the domain to which a computer belongs can be determined in a straightforward manner from the computer's IP address by a "reverse DNS lookup," a network service set up to provide precisely this type of information. Thus, given the router-level network topology, it is a simple task to determine the domain to which each router belongs and group vertices in the network according to their domain. An edge is then placed between two vertices if any router in one has a direct network connection to any router in the other. The study by Faloutsos *et al.* [111] mentioned earlier looked at the domain-level structure of the Internet as well as the router-level structure.

The third common coarse-graining of the network is coarse-graining at the level of autonomous systems. An autonomous system is similar to a domain: it is a group of computers, usually under single administrative control, and it often (though not always) coincides with a domain. Coarse-graining at the autonomous system level is not usually used with data derived from traceroute sampling but with data derived using an alternative method based on BGP routing tables, for which it forms the most natural unit of representation. The BGP method and autonomous systems are discussed in detail in the next section.

2.1.2 MEASURING INTERNET STRUCTURE USING ROUTING TABLES

Internet routers maintain *routing tables* that allow them to decide in which direction incoming packets should be sent to best reach their destination. Routing tables are constructed from information shared between routers using the Border Gateway Protocol (BGP). They consist of lists of complete paths from the router in question to destinations on the Internet. When a packet arrives at a router, the router examines it to determine its destination and looks up that destination in the routing table. The first step of the path in the appropriate table entry tells the router how the packet should be sent on its way. Indeed, in theory routers need store only the first step on each path in order to route packets correctly. However, for efficient calculation of routes using BGP (the techniques of which we will not go into here) it is highly desirable that routers be aware of the entire path to each destination, and since the earliest days of the Internet all routers have operated in this way. We can make use of this fact to measure the structure of the Internet.

Routing tables in routers are represented at the level of *autonomous systems* (ASes). An autonomous system is a collection of computers and routers, usually under single administrative control, within which data routing is handled independently of the wider Internet, hence the name "autonomous system." That is, when a data packet arrives at a router within an autonomous system, destined for a specific computer within that same autonomous system, it is the responsibility of the autonomous system to get the packet the last few steps to its final destination. Data passing between autonomous systems, however, is handled by the Internet-wide mechanisms of BGP. Thus it's necessary for BGP to know about routing only down to the level of autonomous systems and hence BGP tables are most conveniently represented in autonomous system terms. In practice, autonomous systems, of which there are (at the time of writing) about twenty thousand on the Internet, often coincide with domains, or nearly so.

Autonomous systems are assigned unique identification numbers. A routing path consists of a sequence of these AS numbers and since router tables consist of paths to a large number of destinations, we can construct a picture of the Internet at the autonomous system level by examining them. The process is very similar to that used for the traceroute method described in the previous section and depicted in Fig. 2.2. We first obtain a number of router tables. This is normally done simply by the gracious cooperation of router operators at a variety of organizations. Each router table contains a large number of paths starting from a single source (the router), and the union of these paths gives a good but not complete network snapshot in which the vertices are autonomous systems and the edges are the connections between autonomous systems. As with traceroute, it is important that the routers used be well scattered over the network to avoid too much duplication of results, and the number of routers used should be as large as possible to make the sampling of network edges as complete as possible. For example, the Routeviews Project,[8] a large BGP-based Internet mapping effort based at the University of Oregon, uses (again at the time of writing) a total of 223 source computers around the world to measure the structure of the entire network every two hours.

Figure 2.3 shows a picture of the Internet at the AS level derived from routing tables. Qualitatively, the picture is similar to Fig. 1.1 for the class C subnet structure, but there are differences arising because class C subnets are smaller units than many autonomous systems and so Fig. 1.1 is effectively a finer-grained representation than Fig. 2.3.

Using router-, subnet-, domain-, or AS-level structural data for the Internet, many intriguing features of the net's topology have been discovered in recent years [57, 66, 111, 211, 262, 265], many of which are discussed in later chapters of this book.

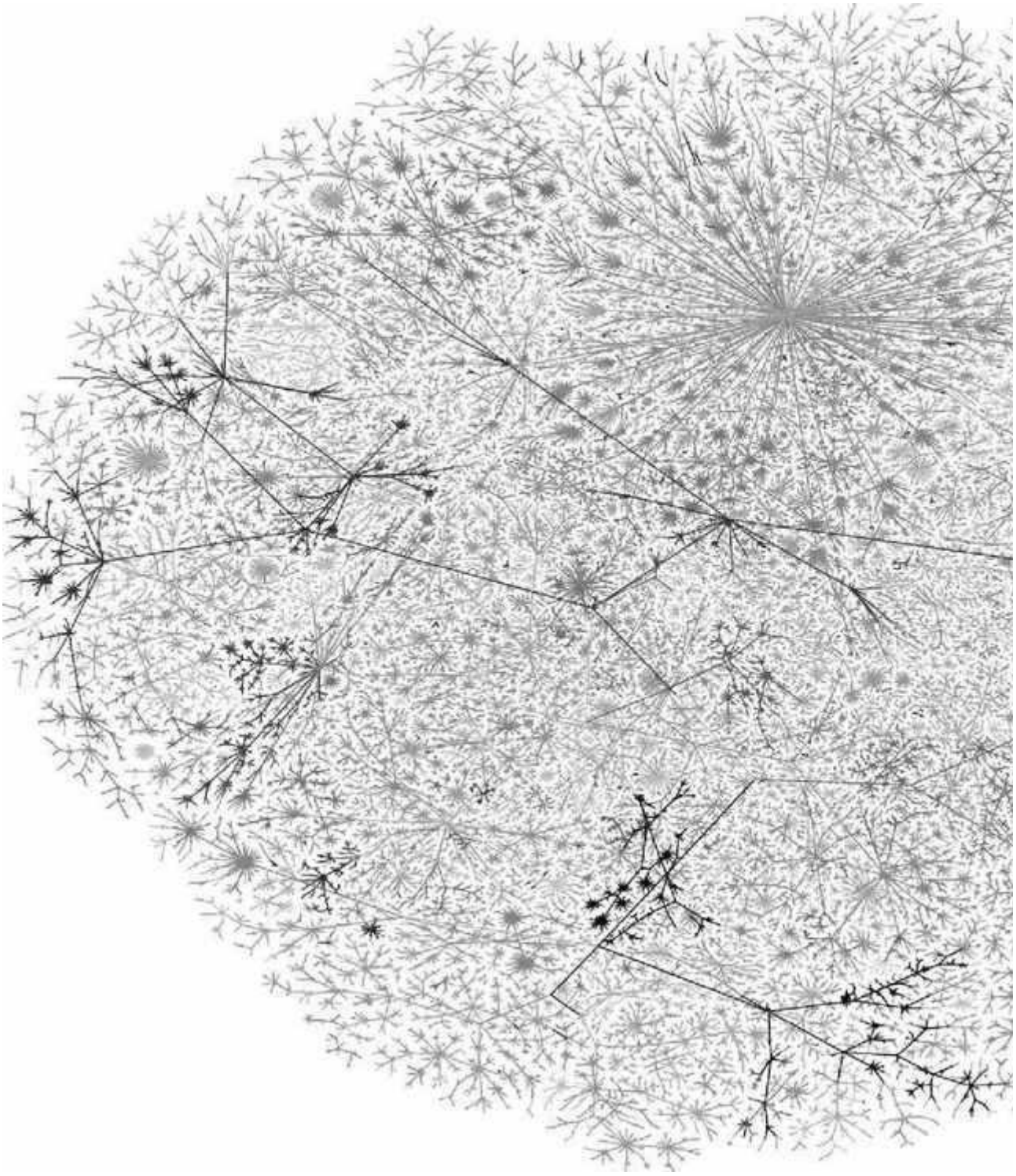


Figure 2.3: The structure of the Internet at the level of autonomous systems. (See Plate III for color version.) The vertices in this network representation of the Internet are autonomous systems and the edges show the routes taken by data traveling between them. This figure is different from Fig. 1.1, which shows the network at the level of class C subnets. The picture was created by Hal Burch and Bill Cheswick. Patent(s) pending and Copyright Lumeta Corporation 2009. Reproduced with permission.

One further aspect of the Internet worth mentioning here is the geographic location of its vertices on the surface of the Earth. In many of the networks that we will study in this book, vertices do not exist at any particular position in real space—the vertices of a citation network for instance are not located on any particular continent or in any particular town. Not so the Internet; its vertices, by and large, are quite well localized in space. Your computer sits on your desk, a router sits in the basement of an office building, and so forth. Things become more blurry once the network is coarse-grained. The domain umich. edu covers large parts of the state of Michigan. The domain aol. com covers most of North America. These are somewhat special cases, however, being unusually large domains. The majority of domains have a well-defined location at least to within a few miles. Furthermore, tools now exist for determining, at least approximately, the geographic location of a given IP address, domain, or autonomous system. Examples include *NetGeo*, *NetAcuity*, *GeoNetMap*, and many others. Geographic locations are determined primarily by looking them up in one of several registries that record the official addresses of the registered owners of domains or autonomous systems. These addresses need not in all cases correspond to the actual location of the corresponding computer hardware. For instance, the domain ibm. com is registered in New York City, but IBM's principal operations are in California. Nonetheless, an approximate picture of the geographic distribution of the Internet can be derived by these methods, and there has been some interest in the results [332].

Geographic localization is a feature the Internet shares with several other technological networks, as we will see in the following sections, but rarely with networks of other kinds.[9]

2.2 THE TELEPHONE NETWORK

The Internet is the best studied example of a technological network, at least as measured by volume of recent academic work. This is partly because data on Internet structure are relatively easy to come by and partly because of intense interest among engineers and computer scientists and among the public at large. Several other technological networks however are worthy of mention here. In this and the following sections of the chapter we look briefly at the telephone network and various distribution and transportation networks. A few other networks, such as software call graphs and electronic circuits, could also be considered technological networks and have been studied occasionally, but are beyond the scope of this book.

The telephone network—meaning the network of landlines and wireless links[10] that transmits telephone calls—is one of the oldest communication networks still in use (although the postal network is certainly older), but it has been little studied by network theorists, primarily because of a lack of good data about its structure. Of course, the structure of the phone network is known, but the data are largely proprietary to the telephone companies that own the network and, while not precisely secret, they are not openly shared with the research community in the same way that Internet data are. We hope that this situation will change, although the issue may become moot in the not too distant future, as telephone companies are sending an increasing amount of voice traffic over the Internet rather than over dedicated telephone lines, and it may not be long before the two networks merge into one.

Some general principles of operation of the telephone network are clear however. By contrast with the Internet, the traditional telephone network is, as mentioned in Section 2.1, not packet switched. Signals sent over the phone network are not disassembled and sent as sets of discrete packets. Instead the telephone network is *circuit switched*, which means that the telephone company has a number of lines or circuits available to carry telephone calls between different points and it assigns them to individual callers when those callers place phone calls. In the earliest days of the telephone systems in the United States and Europe the "lines" actually were individual wires, one each for each call the company could carry. Increasing the capacity of the network to carry more calls meant putting in more wires. Since the early part of the twentieth century, however, phone companies have employed techniques for *multiplexing* phone signals, i.e., sending many calls down the same wire simultaneously. The exception is the "last mile" of connection to the individual subscriber. The phone cable entering a house usually only carries one phone call at a time, although even that has changed in recent years as new technology has made it possible for households to have more than one telephone number and place more than one call at a time.

The basic form of the telephone network is relatively simple. Most countries with a mature landline (as opposed to wireless) telephone network use a three-tiered design. Individual telephone subscribers are connected over local lines to local telephone exchanges, which are then connected over shared "trunk" lines to long-distance offices, sometimes also called toll-switching offices. The long-distance offices are then connected among themselves by further trunk lines. See Fig. 2.4 for a sketch of the network structure. The structure is, in many ways, rather similar to that of the Internet (Fig. 2.1), even though the underlying principles on which the two networks operate are quite different.

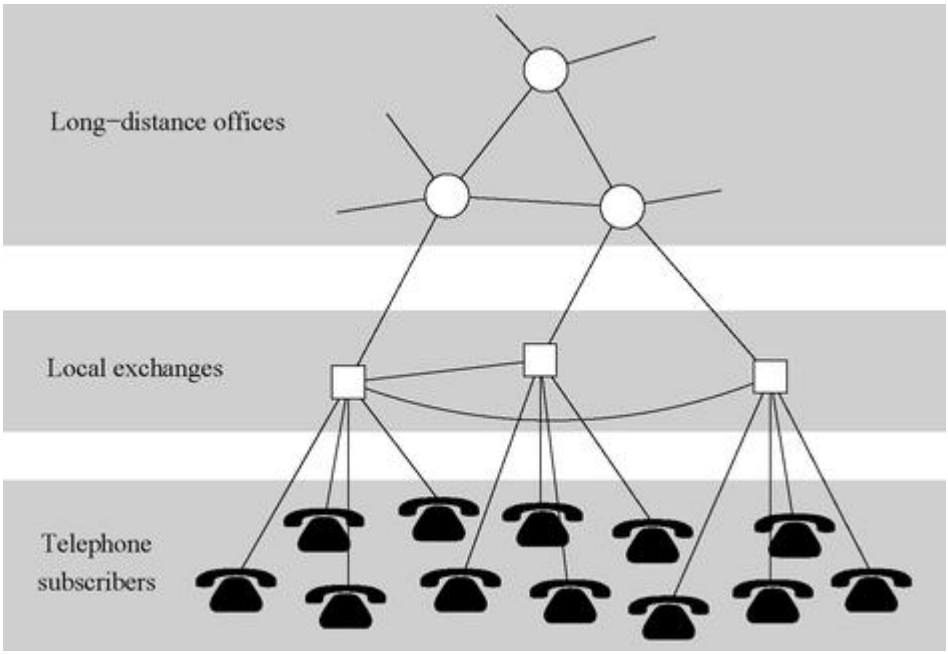


Figure 2.4: A sketch of the three-tiered structure of a traditional telephone network. In a telephone network individual subscriber telephones are connected to local exchanges, which are connected in turn to long-distance offices. The long-distance offices are connected amongst themselves by further lines, and there may be some connections between local exchanges as well.

The three-level topology of the phone network is designed to exploit the fact that most telephone calls in most countries are local, meaning they connect subscribers in the same town or region. Phone calls between subscribers connected to the same local exchange can be handled by that exchange alone and do not need to make use of any trunk lines at all. Such calls are usually referred to as local calls, while calls that pass over trunk lines are referred to as trunk or long-distance calls. In many cases there may also be direct connections between nearby local exchanges that allow calls to be handled locally even when two subscribers are not technically attached to the same exchange.

The telephone network has had roughly this same topology for most of the last hundred years and still has it today, but many of the details about how the network works have changed. In particular, at the trunk level some telephone networks are no longer circuit switched. Instead they are now digital packet switched networks that work in a manner not dissimilar from the Internet, with voice calls digitized, broken into packets, and transmitted over optical fiber links. Only the "last mile" to the subscriber's telephone is still carried on an old-fashioned dedicated circuit, and even that is changing with the advent of digital and Internet telephone services. Nonetheless, in terms of geometry and topology the structure of the phone network is much the same as it has always been, being dictated in large part by the constraints of geography and the propensity for people to talk more often to others in their geographic vicinity than to those further away.

2.3 POWER GRIDS

The topology of power grids has received occasional study in the networks literature [16, 323]. A power grid, in this context, is the network of high-voltage transmission lines that provide long-distance transport of electric power within and between countries. Low-voltage local power delivery lines are normally excluded. The vertices in a power grid correspond to generating stations and switching substations, and the edges correspond to the high-voltage lines. The topology of power grids is not difficult to determine. The networks are usually overseen by a single authority and complete maps of grids are readily available. Indeed, very comprehensive data on power grids (as well as other energy-related networks such as oil and gas pipelines) are available from specialist publishers, either on paper or in electronic form, if one is willing to pay for them.

There is much of interest to be learned by looking at the structure of power grids. Like the Internet, power grids have a spatial aspect; the individual vertices each have a location somewhere on the globe, and their distribution in space is interesting from geographic, social, and economic points of view. Network statistics, both geographic and topological, may provide insight into the global constraints governing the shape and growth of grids. Power grids also display some unusual behaviors, such as cascading failures, which can give rise to surprising results such as the observed power-law distribution in the sizes of power outages [92].

However, while there is a temptation to apply simple models of the kind described in this book to try to explain these and other results, it is wise to be cautious. Power grids are very complicated systems. The flow of power is governed not only by simple physical laws, but also by precise and detailed control of the phases and voltages across transmission lines, monitored and adjusted on rapid timescales by sophisticated computer systems and on slower timescales by human operators. It turns out that power failures and other power-grid phenomena are influenced relatively little by the raw topology of the network and much more by operator actions and software design, and as a result network theory has not, so far, been very successful at shedding light on the behavior of power grids.

2.4 TRANSPORTATION NETWORKS

A moderate amount of work has been done on the structure and function of transportation networks such as airline routes and road and rail networks. The structure of these networks is not usually hard to determine, although compiling the data may be laborious. Airline networks can be reconstructed from published airline timetables, road and rail networks from maps. Geographic information systems (GIS) software can be useful for speeding the compilation of transportation data, and there are also a variety of online resources providing useful information such as latitude and longitude of airports.

One of the earliest examples of a study of a transportation network is the study by Pitts [268] of waterborne transport on Russian rivers in the Middle Ages. There was also a movement among geographers in the 1960s and 70s to study road and rail networks, particularly focusing on the interplay between their economics and their physical structure. The most prominent name in the movement was that of Karel Kansky, and his book on transportation networks is a good point of entry into that body of literature [168].

More recently a number of authors have produced studies applying new network analysis ideas to road, rail, and air networks [16, 136, 294]. In most of the networks studied the vertices represent geographic locations and the edges routes between them. For instance, in studies of road networks the vertices usually represent road intersections and the edges roads. The study by Sen *et al.* [294] of the rail network of India provides an interesting counterexample. Sen *et al.* argue, plausibly, that in the context of rail travel what matters to most people is whether there is a direct train to their destination or, if there is not, how many trains they will have to take to get there. People do not care so much about how many stops there are along the way, so long as they don't have to change trains. Thus, Sen *et al.* argue, a useful network representation in the case of rail travel is one in which the vertices represent locations and two vertices are connected by an edge if a single train runs between them. Then the distance between two vertices in the network—the number of edges you need to traverse to get from A to B—is equal to the number of trains you would have to take. A better representation still (although Sen *et al.* did not consider it) would be a "bipartite network," a network containing two types of vertex, one representing the locations and the other representing train routes. Edges in the network would then join locations to the routes that run through them. The first, simpler representation of Sen *et al.* can be derived from the bipartite one by making a "one-mode projection" onto the locations only. Bipartite networks and their projections are discussed in greater detail in Section 6.6.

2.5 DELIVERY AND DISTRIBUTION NETWORKS

Falling somewhere between transportation networks and power grids are the distribution networks, about which relatively little has been written within the field of networks research. Distribution networks include things like oil and gas pipelines, water and sewerage lines, and the routes used by the post office and package delivery and cargo companies. Figure 2.5 shows one example, the European gas distribution network, taken from a study by Carvalho *et al.* [64], who constructed the figure from data purchased from industry sources. In this network the edges are gas pipelines and the vertices are their intersections, including pumping, switching, and storage facilities and refineries.

If one is willing to interpret "distribution" in a loose sense, then one class of distribution networks that has been relatively well studied is river networks, though if one wants to be precise river networks are really collection networks, rather than distribution networks. In a river network the edges are rivers or streams and the vertices are their intersections. Like road networks no special techniques are necessary to gather data on the structure of river networks—the hard work of surveying the land has already been done for us by surveyors and cartographers, and all we need do is copy the results off their maps. See Fig. 2.6 for an example of a river network.

The topological and geographic properties of river networks have been studied in some detail [94, 208, 284]. Of particular note is the fact that river networks, to an excellent approximation, take the form of trees. That is, they contain no loops (if one disregards the occasional island midstream), a point that we discuss in more detail in Section 6.7.

Similar in some respects to river networks are networks of blood vessels in animals, and their equivalents in plants, such as root networks. These too have been studied at some length. An early example of a mathematical result in this area is the formula for estimating the total geometric length of all edges in such a network by observing the number of times they intersect a regular array of straight lines [231]. This formula, whose derivation is related to the well-known "Buffon's needle" experiment for determining the value of π, is most often applied to root systems, but there is no reason it could not also be useful in the study of river networks or, with suitable modification, any other type of geographic network.

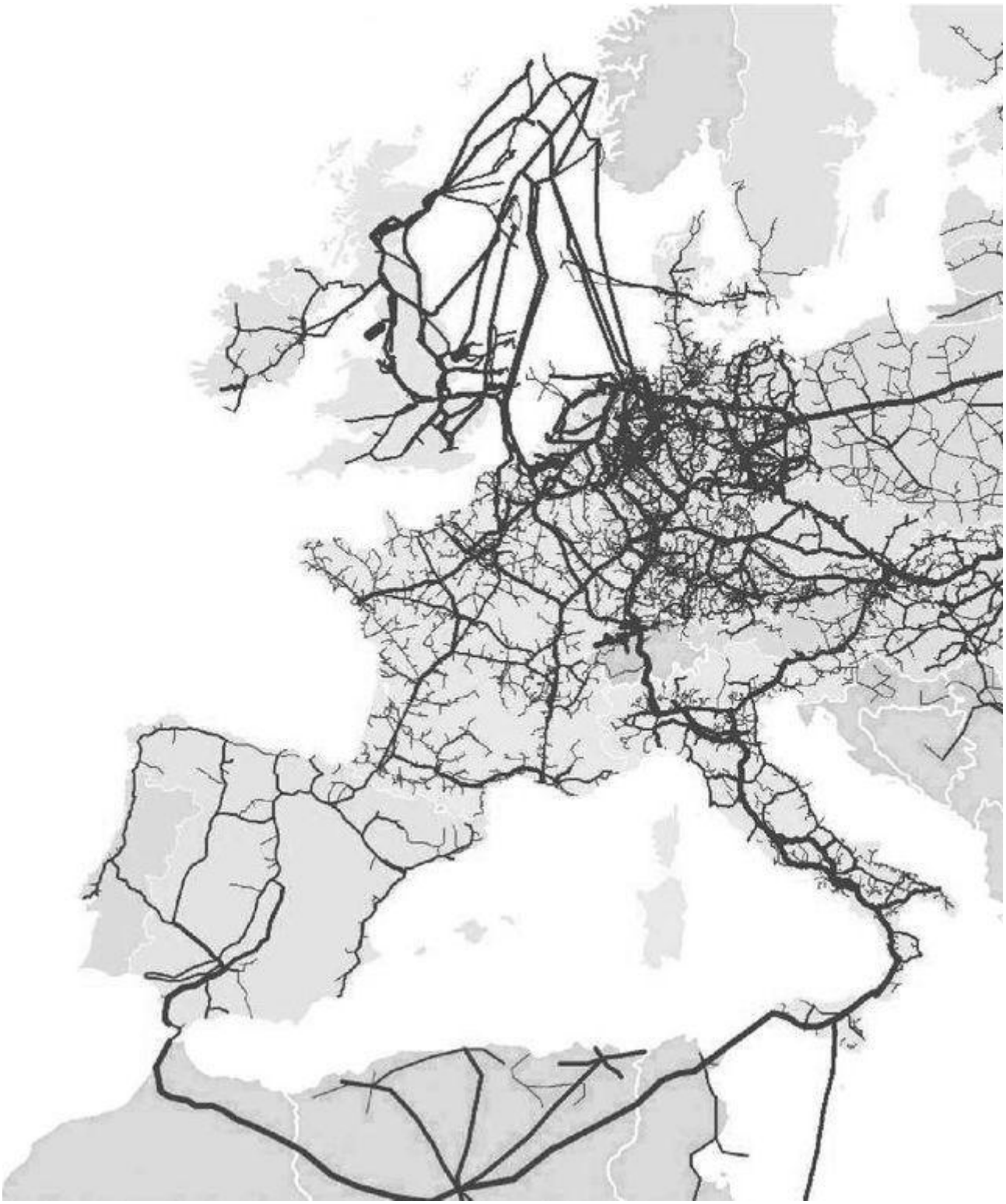


Figure 2.5: The network of natural gas pipelines in Europe. Thickness of lines indicates the sizes of the pipes. Figure created by R. Carvalho *et al.* [64]. Copyright 2009 American Physical Society. Reproduced with permission.

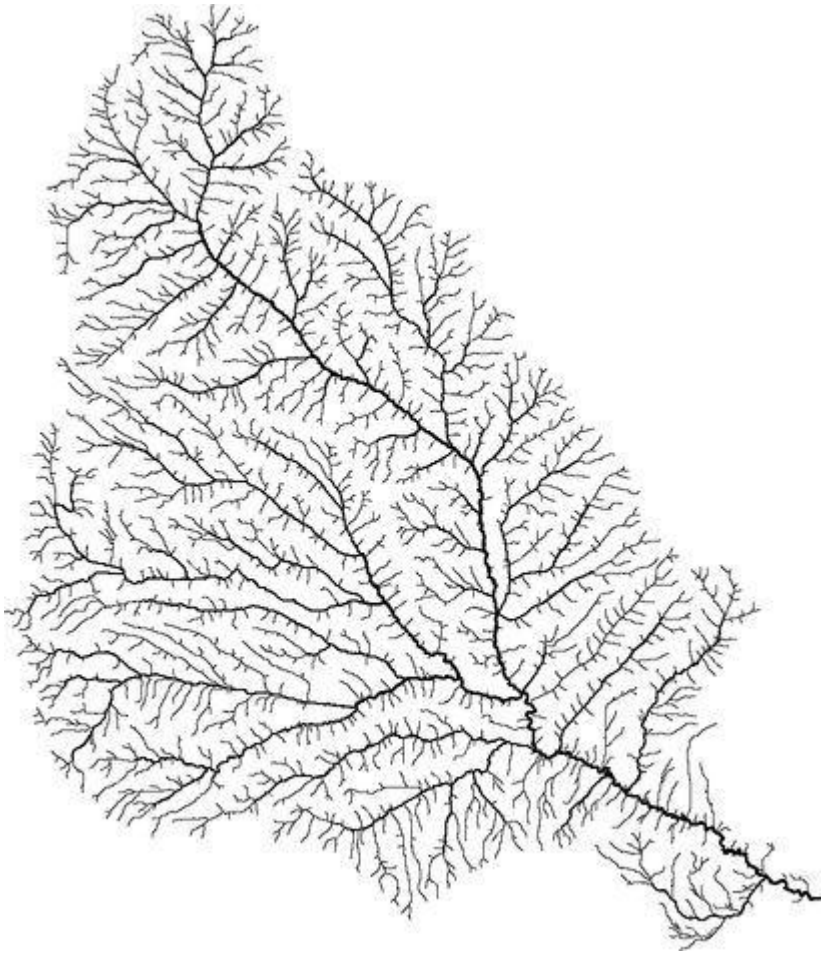


Figure 2.6: Drainage basin of the Loess Plateau. The network of rivers and streams on the Loess Plateau in the Shanxi province of China. The tree-like structure of the network is clearly visible—there are no loops in the network, so water at any point in the network drains off the plateau via a single path. Reproduced from Pelletier [266] by permission of the American Geophysical Union.

Also of note in this area is work on the scaling relationships between the structure of branching vascular networks in organisms and metabolic processes [26, 325, 326], an impressive example of the way in which an understanding of network structure can be parlayed into an understanding of the functioning of the systems the networks represent. We will see many more examples during the course of this book.